



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 9

TASKS :

1. Declare a 1D array called `dailySteps` with 7 elements to store your step count for each day of the week, assign sample values, and print each value using a loop.



```
#include <stdio.h>

int main()
{
    int dailySteps[7] = {5000, 6500, 7200, 8000, 9000, 7500, 10000};
    int i;

    printf("Daily Steps:\n");

    for(i = 0; i < 7; i++)
    {
        printf("Day %d: %d\n", i + 1, dailySteps[i]);
    }

    return 0;
}
```

OUTPUT:

```
Daily Steps:
Day 1: 5000
Day 2: 6500
Day 3: 7200
Day 4: 8000
Day 5: 9000
Day 6: 7500
Day 7: 10000
```

2. Create a 2D array called `playlistRatings` to store ratings for 3 Spotify playlists over 5 days (rows = playlists, columns = days). Fill it with sample numbers and print the ratings for the second playlist.



```
#include <stdio.h>

int main()
{
    int playlistRatings[3][5] =
    {
        {5, 4, 5, 4, 5},
        {3, 4, 5, 5, 4},
        {4, 5, 4, 5, 5}
    };

    int i;

    printf("Ratings for Second Playlist:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d ", playlistRatings[1][i]);
    }

    return 0;
}
```

OUTPUT:

```
Ratings for Second Playlist:
3 4 5 5 4
```

3. Build a function that takes a 1D array of 7 integers representing your daily Zomato order amounts and calculates the average spend for the week.

Hint: Use a loop to sum the values, then divide by the array length.



```
#include <stdio.h>

float calculateAverage(int orders[])
{
    int i, sum = 0;

    for(i = 0; i < 7; i++)
    {
        sum += orders[i];
    }

    return (float)sum / 7;
}

int main()
{
    int orders[7] = {250, 180, 320, 150, 400, 275, 225};

    printf("\nAverage Weekly Spend = %.2f\n", calculateAverage(orders));

    return 0;
}
```

OUTPUT:

```
Average Weekly Spend = 257.14
```

4. Given a 2D array called `cricketScores` where each row represents an IPL match and columns represent runs scored by each team, write code to print the highest score from each match.



```
#include <stdio.h>

int main()
{
    int cricketScores[3][2] =
    {
        {180, 175},
        {210, 195},
        {165, 170}
    };

    int i;

    printf("Highest Score from Each Match:\n");

    for(i = 0; i < 3; i++)
    {
        if(cricketScores[i][0] > cricketScores[i][1])
        {
            printf("Match %d: %d\n", i + 1, cricketScores[i][0]);
        }
        else
        {
            printf("Match %d: %d\n", i + 1, cricketScores[i][1]);
        }
    }

    return 0;
}
```

OUTPUT:

```
Highest Score from Each Match:
Match 1: 180
Match 2: 210
Match 3: 170
```